

Improving Classification Accuracy For Diabetic Retinopathy With Deep Learning

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Abstract— Diabetic retinopathy (DR) is one of the primary contributors to vision impairment worldwide, highlighting the utmost need for early detection and diagnosis to prevent irreversible damage. In the last decade, deep learning (DL) has had a game-changing impact on medical imaging towards the development of faster and more accurate softwares for disease classification problems like DR detection. Although this has led to several promising achievements, existing models related to DR classification have fundamental limitations such as weak generalization across various datasets and suboptimal performance with real-world data. In this work, the classification of DR is conducted using a new approach combining SqueezeNet with Squeeze-and-Excitation (SE) blocks. The aim of this architecture is to provide improved computational efficiency and better feature extraction compared to more traditional models. The proposed model attained a test accuracy of 90.9% which is substantially better than the previously established results of DR image classification, as evidenced by experiments. The results underscore the potential of leveraging advanced DL techniques to enhance diagnostic accuracy, that may allow for similar clinical uses in the early detection of diabetic retinopathy.

Keywords— Diabetic Retinopathy (DR), Deep Learning (DL), Convolutional Neural Networks (CNN), Image Classification.

I. INTRODUCTION

The most common retinal disorder that develops in individuals with longstanding diabetes, typically due to damage and occlusion of the blood vessels within the retina secondary to high levels of sugar is Diabetic retinopathy (DR).[1] As a result, hemorrhages, microaneurysms, exudates, and cotton-wool spots manifest.[2] Microaneurysms, retinal hemorrhages, exudates, and cotton-wool spots are progressive retinal signs of diabetic retinopathy, caused by fluid and blood leakage from weakened vessels, indicating escalating risk of vision loss.[3] Diabetic retinopathy is a leading cause of blindness, requires early diagnosis, which has driven the development of automated systems due to the increasing number of patients and the limited availability of skilled technicians.[4] Diabetic retinopathy (DR) is a slowly progressive of the retina that results from prolonged high blood glucose and blood pressure in diabetic individuals, making it a common cause of vision loss worldwide.[5] Approximately one-third of diabetic patients develop DR, which damages the retinal microvasculature and leads to abnormalities like microaneurysms, haemorrhages, exudates, and capillary closures, ultimately causing vision loss.[6] The World Health Organization (WHO) projects diabetes to become the 7th leading cause of death, with the global diabetic population rising from 108 million in 1980 to 422 million by 2014,

particularly affecting those below the poverty line, and India alone is expected to see an increase from 61.3 million diabetic patients in 2023 to 101.2 million by 2030.[7]

The disease progresses through five stages, from no retinopathy to proliferative DR, with increasing severity marked by lesions, haemorrhages, and neovascularization, necessitating prompt treatment to prevent further damage.[8] The severity of diabetic retinopathy is determined by the presence and extent of retinal abnormalities. Key diagnostic indicators include microaneurysms, haemorrhages, neovascularization, and venous beading. Microaneurysms are small blood clots, 100–120 μm in size, while haemorrhages result from blood vessel rupture.[9] Neovascularization refers to abnormal new blood vessel growth, and venous beading indicates the dilation of veins near occluded arterioles. Based on disease progression, Diabetic retinopathy is classified into Non-Proliferative Diabetic Retinopathy (NPDR) and Proliferative Diabetic Retinopathy (PDR), with NPDR further divided into mild, moderate, and severe stages.[10]

Early detection of diabetic retinopathy (DR) is paramount, as timely intervention can prevent the visual impairment and blindness associated with the disease.[11] Despite DR's progressive nature, its debilitating effects can be mitigated with prompt diagnosis and treatment. Regular digital fundus imaging plays a vital role in identifying early signs of DR, allowing for immediate management before irreversible damage occurs. Clinical guidelines stress the importance of lifelong, routine screening for diabetic patients, with more frequent examinations recommended for those in advanced stages of the disease.[12] However, the rising global prevalence of DR and the strain on healthcare systems underscore the necessity of efficient early detection methods to alleviate the burden on clinicians and ensure effective patient care.[13]

Conventional methods for diagnosing diabetic retinopathy (DR) typically rely on human evaluation of retinal fundus images by expert ophthalmologists or employ image processing techniques that use handcrafted features to identify abnormalities.[14] These methods, though effective to a certain extent, are limited by their dependence on human expertise and the difficulty of accurately capturing the complex and subtle patterns associated with different stages of diabetic retinopathy.[15] Furthermore, the variability in image quality, patient diversity, and disease progression presents challenges for traditional approaches, often resulting in inconsistent and delayed diagnoses.[16]

Deep learning has become a powerful technique for processing complex medical datasets, particularly in the realm of medical imaging.[17] These models can extract subtle features automatically across different imaging

modalities with high accuracy and efficiency using the convolutional neural network. Deep learning has advantages especially in the context of diabetic retinopathy (DR), as it enables automated detection these critical retinal abnormalities, such as microaneurysms, exudates and haemorrhages.[18] This capability not only facilitates early diagnosis and precise monitoring of disease progression but also provides a scalable solution for mass screening, especially in underserved regions with limited access to specialised medical expertise. As deep learning technology develops further, clinical application of this shows promise in revolutionising trend like detection and management of DR with improved patient outcomes at increased access to effective treatment.[19]

Shanthi and Sabeenian[8] conducted a study focused on diabetic retinopathy classification utilized a modified AlexNet CNN to categorize fundus images into disease stages, achieving over 95% accuracy on the Messidor dataset and highlighting potential improvements with larger datasets.

Naz and Ahuja[20] developed RetinaCare, a web-based tool leveraging VGG-16 CNN for automated diabetic retinopathy (DR) grading. Trained on IDRiD and Messidor datasets, it achieved 96.89% accuracy, demonstrating robust diagnostic potential. Future efforts aim to validate performance in real-world remote settings.

Previous models for diabetic retinopathy (DR) classification often fall short in balancing accuracy and computational efficiency, especially when detecting early-stage signs such as microaneurysms and exudates. Many of these models rely on complex architectures or handcrafted features, which can be computationally intensive and struggle to generalize across diverse datasets, leading to inconsistent results. This paper presents a novel approach using SqueezeNet with a Squeeze-and-Excitation (SE) block, designed to address these challenges. The combination of SqueezeNet's lightweight architecture and the SE block's enhanced feature recalibration improves both accuracy and efficiency, making the model well-suited for scalable applications in clinical settings. The objective of this model is to deliver a more resource-efficient yet highly precise tool for the early detection and classification of DR, thereby improving diagnostic outcomes and accessibility to automated screening.

The paper is organized as follows: it begins with a Literature Review, followed by an outline of the Proposed Methodology. This is trailed by the Experimental Results and Analysis, which includes subsections on Dataset Description, Implementation Details, and Results. Finally, the paper concludes with a comprehensive summary and a discussion of future work.

II. LITERATURE REVIEW

Deep learning (DL) is increasingly utilised across various medical fields because of its ability to automatically excerpt complex features from medical imaging data, improving tasks such as tumour detection, organ segmentation, and disease classification. Its success in analysing large-scale datasets makes it highly effective in areas like radiology, dermatology, and pathology. In the context of diabetic retinopathy (DR), DL has become crucial for early identification by recognizing significant retinal abnormalities, such as exudates and microaneurysms, from

fundus images. This capability enables DL models to provide more accurate, scalable, and efficient solutions for DR screening and diagnosis, addressing the need for automated tools in healthcare.

Teng et. al.,[21] MediDRNet, a prototypical contrastive learning based diabetic retinopathy (DR) classifier utilizing dual-branch network model, addressing sample imbalance by focusing on lesion differences and similarities. The model's dual-branch structure transitions from feature learning, constructing a hierarchical system for lesion progression, to classification learning. To enable the identification of subtle retinal features, MediDRNet involves convolutional block attention module (CBAM) to improve spatial and channel-level attention. Experiments on Kaggle and UWF datasets showed that our approach achieved state of the art results, especially for minority DR categories.

In the paper[22], Madarapu et. al. proposed the MuR-CAN model for diabetic retinopathy (DR) classification, addressing challenges like inter-class variances and minor retinal lesions. A multi-dilation attention block (MDAB) is incorporated into the model, using depth-wise convolutions with varying dilation rates to capture multi-scale spatial information, enhancing focus on key features. Classification is carried out using a support vector machine (SVM), which increases accuracy. State-of-the-art methods are significantly outpaced by MuR-CAN, which combines CNNs like Xception and ResNet50V2. Future work aims to tackle data imbalance using advanced augmentation techniques to enhance generalization and classification outcomes.

Abini and Priya's[23] study employed deep learning to classify diabetic retinopathy (DR) across normal, mild, moderate, severe, and proliferative stages using pre-trained CNNs, VGG-16, and MobileNet-V2. Aiming to overcome the limitations of manual feature extraction in previous research, the authors automated the classification process, achieving 90% and 92% accuracy in estimating DR stages. This approach effectively aids ophthalmologists in early DR detection, crucial for preventing severe visual impairments. The integration of advanced CNN architectures demonstrated significant advancements in DR diagnosis, enhancing diagnostic efficiency in clinical settings.

In a recent study[1], Madarapu et. al. developed a novel deep integrative tactic for classifying diabetic retinopathy (DR) by addressing challenges like imbalanced datasets and small lesions while offering greater inter-class similarity. Using residual blocks with a channel-spatial attention mechanism (CSAM), the model captures local and global information, while non-local blocks (NLB) further enhance important features and suppress irrelevant ones. This combination refines outputs and improves understanding of complex image relationships, leading to superior classification accuracy. The model also achieves efficiency with fewer trainable parameters and faster inference times than many state-of-the-art techniques. Experimental results show improved accuracy on comprehensive DR datasets.

III. METHODOLOGY

The integration of computer-aided technologies has revolutionized healthcare, improving diagnostic accuracy and streamlining data management through systems like CAD and electronic health records. Telemedicine has expanded access to care, while automated image analysis in

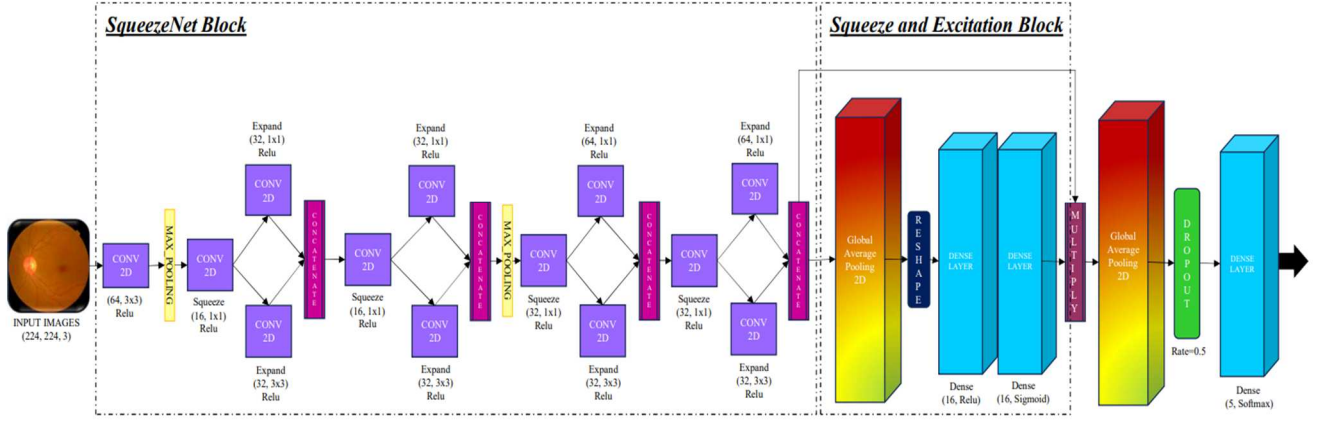


Fig. 1. The architecture of SqueezeNet with Squeeze and Excitation Block

diabetic retinopathy (DR) screening enhances accuracy and enables timely interventions, leading to better patient outcomes and more effective diabetes management.

Deep learning (DL) and artificial intelligence (AI) have significantly advanced medical diagnostics, particularly for DR detection.[24] Using convolutional neural networks (CNNs), DL algorithms analyse retinal images with exceptional precision, identifying subtle features often missed by human evaluators. Transfer learning further optimizes this process by fine-tuning pre-trained models for specific datasets, addressing the challenge of limited labelled medical data. These technologies improve diagnostic accuracy, reduce human error, conserve resources, and facilitate the development of efficient, reliable diagnostic tools, transforming DR management and making it more accessible and effective. The model employed in this research is based on SqueezeNet which is a lightweight convolutional neural network architecture that achieves greater accuracy with fewer parameters. By using a fire module, SqueezeNet compresses the number of input channels before applying filters, enabling efficient computation without sacrificing performance. This design is particularly advantageous in environments with limited computational resources, such as mobile devices or edge computing applications. To further enhance the model's performance, a Squeeze-and-Excitation (SE) block is integrated with the SqueezeNet architecture. The SE block introduces a mechanism for channel-wise attention, allowing the network to focus on the most informative features while suppressing less relevant ones. This is achieved by first squeezing the feature maps to produce a channel descriptor, that goes through a learnable activation function. The resultant weights are multiplied back to the original feature maps, effectively enhancing the network's ability to capture important patterns in the data. A sneak peek of the model's architecture is represented in Fig. 1. The combination of SqueezeNet and the SE block offers several key benefits. Firstly, the model maintains a compact size, making it suitable for deployment in environments with limited resources while still delivering competitive accuracy. Secondly, the incorporation of the SE block improves feature representation by emphasizing the most critical channels, resulting in more robust classification performance. As a result, this model is well-suited for tasks such as diabetic retinopathy detection, where the ability to discern subtle

variations in retinal images is crucial. By leveraging the strengths of SqueezeNet and the SE block, this approach provides an effective solution for the efficient classification of medical images, ultimately contributing to better patient care and outcomes. The SqueezeNet model processes input images through a series of well-defined layers to evaluate and classify them efficiently. Initially, the model accepts a 224×224 -pixel RGB image, which undergoes a 3×3 convolutional layer with ReLU activation to extract low-level features. A max pooling layer follows, condensing spatial dimensions while retaining critical statistics. The core of the architecture consists of multiple fire modules, where each module applies a squeeze layer (a 1×1 convolution) to compress the feature maps, followed by an expand layer that uses both 1×1 and 3×3 convolutions to capture diverse features. After processing through these modules, a Squeeze-and-Excitation block enhances feature representation by applying global average pooling to compute channel-wise attention, letting the model to concentrate on the utmost informative features. Finally, a global average pooling layer reduces the dimensionality of the feature maps to a single vector, followed by dropout for regularization, and concludes with a softmax output layer that classifies the input image into one of five target classes. This layered approach enables the model to efficiently evaluate and classify medical images, making it particularly suitable for applications such as diabetic retinopathy detection.

The SqueezeNet model with Squeeze-and-Excitation blocks stands out due to its unique architecture that effectively balances computational efficiency and high classification accuracy. Its innovative use of fire modules allows for a significant reduction in the number of parameters while maintaining robust feature extraction capabilities. Additionally, the integration of the Squeeze-and-Excitation block enhances the model's performance by enabling it to focus on the most relevant features, thereby improving the representational power of the learned features. This combination not only leads to faster inference times but also results in improved accuracy, making it particularly effective for complex tasks such as diabetic retinopathy detection. These advantages underscore the decision to adopt this model, as it aligns with the need for a lightweight yet powerful solution in medical image classification, ultimately contributing to better diagnostic outcomes and patient care.

A. Dataset Description

The dataset[25] used in this study consists of 3662 retina scan images, resized to 224x224 pixels for compatibility with pre-trained deep learning models; some sample images are displayed class wise in Fig. 2. Sourced from the APTOS 2019 Blindness Detection challenge, the dataset is well-organized, with images divided into five distinct groups depending on the severity of diabetic retinopathy. These categories include No_DR (1805 images, representing healthy eyes), Mild (370 images, indicating early-stage retinopathy), Moderate (999 images, showing intermediate severity), Severe (193 images, highlighting advanced retinopathy), and Proliferate_DR (295 images, reflecting the most severe stage). The balanced structure and clear categorization of this dataset provides a robust foundation for training models to detect varying stages of diabetic retinopathy, making it an ideal choice for this research.

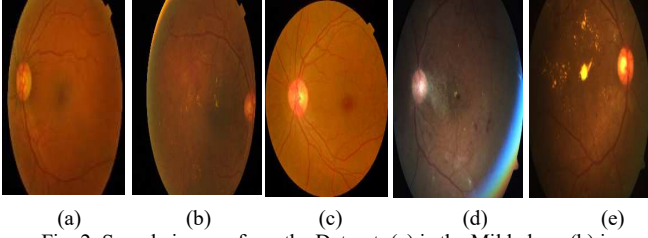


Fig. 2. Sample images from the Dataset: (a) is the Mild class, (b) is Moderate class, (c) is No_DR class, (d) is Proliferative_DR class and (e) is Severe class

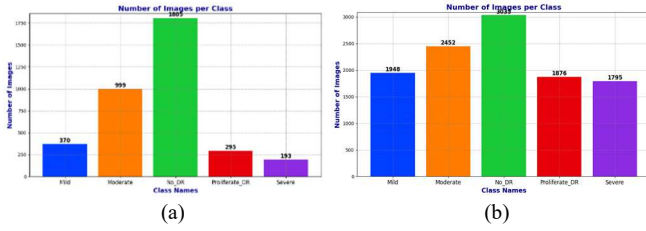


Fig.3. Bar Chart visualization of image dataset Before and After Augmentation

B. Data Preprocessing

In this research, the dataset comprising retina scan images were resized to 224x224 pixels. This resizing ensured compatibility with pre-trained deep learning models, facilitating smoother integration and faster processing. Upon analyzing the dataset, it was evident that there was a significant class imbalance, with certain classes like No_DR being overrepresented compared to others such as Severe and Proliferate_DR. To address this, class balancing techniques were employed to ensure that the model received an equal representation of all severity levels during training. TABLE I displays the distribution of images before augmentation and after augmentation. This step is crucial as it ensures the model to be non-biased towards the majority class, leading to more accurate and generalized predictions. Once the class imbalance was rectified, the dataset was split into training set, validation set, and test set in an 80:10:10 ratio, ensuring a proper distribution for model training, validation, and evaluation. TABLE II depicts the image count in train, validation, and test datasets. Fig. 3 shows the visualization of the same in the form of bar chart. Data augmentation was then applied to the training dataset using a variety of transformations, including width and height shifts (up to 20%), rotation (up to 20 degrees), horizontal flips, shear,

zoom, and nearest neighbor filling. These augmentations helped introduce variability in the dataset and also improved the generalizability of the model across a variety of retinal images. This not only reduced overfitting but also enhanced the robustness of the model, allowing it to perform better on diverse retinal images.

C. Implementation Details

The project utilizes a high-performance system with an NVIDIA RTX 3050 GPU (4GB), a 12th Gen Intel Core i7 processor (2.30 GHz), and 16GB RAM, ensuring efficient handling of deep learning tasks. Specifications are detailed in TABLE III. The programming environment includes VS Code with TensorFlow and Python as core tools. Images are resized to 224x224 pixels to optimize memory while preserving detail for accurate classification. A batch size of 8 ensures efficient GPU use, and the Adam optimizer with a categorical cross-entropy loss function enhances training for multiclass classification. The model is trained for 50 epochs across 5 classes, striking a balance between computational efficiency and performance. TABLE IV lists the hyperparameters.

TABLE I. DISTRIBUTION OF IMAGES IN DATASET

Distribution Before Augmentation		Distribution After Augmentation	
Classes	Number of Images	Classes	Number of Images
Mild	370	Mild	1948
Moderate	999	Moderate	2452
No_DR	1805	No_DR	3035
Proliferative_DR	295	Proliferative_DR	1876
Severe	193	Severe	1795

TABLE II. DISTRIBUTION OF IMAGES FOR TRAINING, TESTING AND VALIDATION

Dataset	Number of Images
Train	2883
Test	119
Validation	1113

TABLE III. SYSTEM SPECIFICATIONS

Name	Parameter
GPU	4GB
Graphics Card	RTX 3050
Processor	12th Gen Intel Core i7
Ram	16GB
Operating System	Windows 11
Architecture	64-bit

TABLE IV. DESCRIPTION OF HYPERPARAMETERS

Hyperparameter	Value
Image Size	224x224
Batch Size	8
Optimizer	Adam
Loss Function	Categorical Cross entropy
Epochs	50
Classes	5

In conclusion, the combination of a carefully curated dataset, effective preprocessing techniques, and an optimized implementation has significantly contributed to the performance of the SqueezeNet model enhanced with Squeeze-and-Excitation blocks. The preprocessing steps,

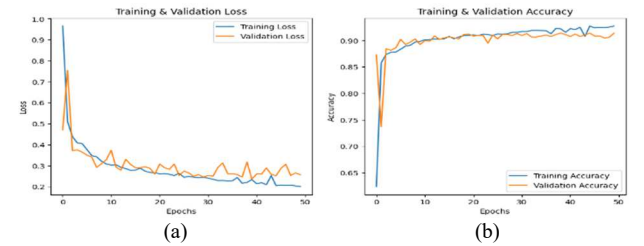


Fig. 4. Performance metrics per epoch illustrate training and validation accuracy and loss

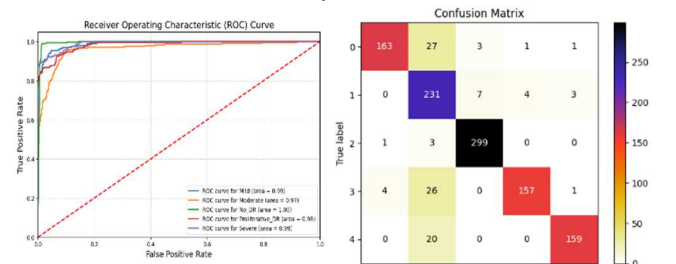


Fig. 5. ROC curve

Fig. 6. Confusion Matrix

including image resizing and class balancing, ensured that the model could handle the data efficiently while maintaining accuracy across all categories. The application of data augmentation further strengthened the generalizability of the model across variety of retinal images simulating various real-world conditions. Additionally, the implementation on a high-performance system allowed for the seamless execution of computationally demanding tasks, enabling the model to achieve superior results in diabetic retinopathy detection. These thoughtful considerations have made the model not only lightweight but also highly effective in delivering accurate and reliable classifications.

IV. EXPERIMENTAL RESULT AND ANALYSIS

The SqueezeNet model enhanced with a Squeeze-and-Excitation (SE) block demonstrated strong performance in the classification of diabetic retinopathy images. The model achieved a training accuracy of 92.72% at the 50th epoch, with a testing accuracy of 90.9% and a validation accuracy of 91.37%. These results underscore the effectiveness of the SE block, that adaptively adjusts channel-wise feature responses, leading to improved model precision. The relatively small size of the SqueezeNet architecture, combined with the SE block's ability to focus on relevant features, provided a good balance between model complexity and performance. This is particularly advantageous for medical image classification tasks where overfitting is a concern, yet high accuracy is critical.

The trends in training and validation loss in Fig. (a) and accuracy in Fig. 4 (b) highlight the model's generalizability and stability. Training and validation loss decreased steadily, with accuracy reaching approximately 90% for both, indicating effective learning and reliability on unseen data. Fig. 5 shows the ROC curves for all classes, with AUC values ranging from 0.97 to 1.00, reflecting excellent classification performance. Notably, the AUC of 1.00 for the No_DR class demonstrates perfect classification. Fig. 6 illustrates the confusion matrix, showing a high number of correct predictions along the diagonal, particularly for No_DR, Proliferative_DR, and Severe, confirming the model's high classification accuracy. Fig. 7 displays the probability distributions, with sharp peaks for No_DR and Severe, reflecting the model's confidence and effective feature focus, supported by the SE block. Fig. 8 presents class-wise precision, recall, and F1-score metrics, with No_DR and Severe achieving particularly high scores, demonstrating strong performance across all classes. The SE block significantly enhances the model's ability to capture subtle class differences, contributing to its robust and accurate classification of diabetic retinopathy. The TABLE V associates the performance of various models on diabetic retinopathy classification in terms of accuracy, precision,

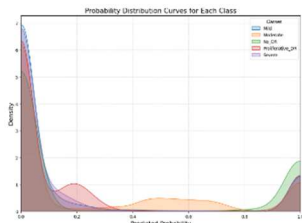


Fig. 7. Probability Distribution Curve

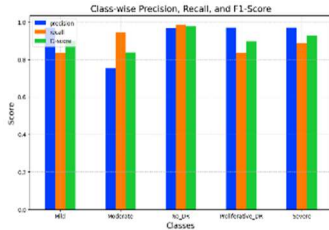


Fig. 8. Class-wise Precision, Recall and F1-Score

recall, and F1-score. Wong et al. achieved 82.10% across all metrics, while Guanghui et al. demonstrated a higher accuracy of 83.40%, though with notably lower precision, recall, and F1-score. Bodapati et al.'s two models show improvements, with the second model achieving 86.22% accuracy, though with slightly lower recall and F1-score. The proposed model outperforms all prior models with a notable 90.90% accuracy, 93.00% precision, and balanced recall and F1-score, highlighting its effectiveness in achieving more accurate and reliable classifications.

TABLE V COMPARISON OF PROPOSED MODEL WITH STATE-OF -THE-ART MODELS

Model	Accuracy	Precision	Recall	F1-Score
Wong et. al.[26]	82.10	82.10	82.10	82.10
Guanghui et. al.[27]	83.40	69.66	67.70	67.02
Bodapati et. al.[28]	84.31	84.00	84.00	84.00
Bodapati et. al.[29]	86.22	80.61	72.23	75.41
Proposed Model	90.90	93.00	90.00	91.00

The model achieved excellent accuracy and minimal loss during training and validation, with clear class distinctions in ROC and probability distributions. Precision-recall metrics and the confusion matrix highlighted its accuracy in identifying critical diabetic retinopathy stages. The Squeeze-and-Excitation (SE) block enhanced feature extraction, ensuring accurate and efficient classification, aligning with the study's objectives.

V. CONCLUSION

In this research, a SqueezeNet model was developed with a Squeeze-and-Excitation (SE) block for the diabetic retinopathy image classification from the APTOS dataset, aiming to enhance diagnostic accuracy and facilitate timely intervention in patients with diabetic retinopathy. The proposed model architecture efficiently balances low computational cost with high performance, leveraging the SE block to emphasize important features while suppressing less informative ones, thereby improving the model's classification capability.

The SqueezeNet with SE block demonstrated impressive efficiency, achieving a test accuracy of 90.9% across five classes: No DR, Mild, Moderate, Severe, and Proliferative DR. This level of performance indicates the model's sturdiness and its potential for deployment in clinical settings, where quick and reliable diagnoses are critical. The lightweight nature of SqueezeNet makes it particularly suitable for deployment on devices with limited computational resources, thus broadening access to advanced diagnostic tools. The successful application of this model highlights its benefits not only in providing accurate classifications but also in reducing the burden on healthcare professionals by automating the screening process. Moreover, this research contributes to the growing body of literature on utilizing deep learning techniques in medical image analysis, emphasizing the need for continued exploration of novel architectures and methods.

Future research directions may include refining the model through data augmentation methods and exploring transfer learning tactics to further enhance performance on smaller

datasets. Additionally, integrating the model with real-time imaging systems could revolutionize diabetic retinopathy screening processes, significantly impacting patient outcomes and healthcare efficiency. The findings of this study underscore the model's potential to transform diabetic retinopathy diagnosis and management, ultimately improving the quality of care for patients at risk.

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